

Academic Organization Ontology

Knowledge Representation Project Report

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Ontology Language:	OWL 2.0
Development Tool:	Protégé 5.x
Reasoner:	HermiT
Repository:	GitHub

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1 Introduction

Knowledge representation is an important area of artificial intelligence that focuses on representing information in a structured and meaningful way. In modern information systems, data is not only stored but also interpreted through relationships, rules, and semantic structures. Ontologies provide a formal method for representing such knowledge.

This project presents an **Academic Organization Ontology** developed using **OWL 2.0** in the **Protégé** environment. The ontology models the structure of an academic institution such as a university. It includes students, professors, departments, faculties, courses, classrooms, libraries, laboratories, examinations, assignments, research centers, scholarships, certificates, and administrative entities.

The main purpose of this ontology is to represent academic knowledge in a machine-understandable form. By defining classes, object properties, data properties, individuals, domains, and ranges, the ontology supports semantic reasoning and information retrieval. The **HermiT Reasoner** was used to verify the logical consistency of the ontology.

2 Background

An ontology is a formal representation of concepts and relationships within a particular domain. It describes the entities that exist in a domain and how they are connected. In Semantic Web technologies, ontologies are commonly written using OWL, the Web Ontology Language.

OWL allows the definition of classes, properties, individuals, and logical restrictions. These components make it possible to reason over data and infer new knowledge. For example, if a person is defined as a student and the student is enrolled in a course, the ontology can answer queries about all enrolled students.

Protégé is a widely used ontology development environment. It provides a graphical interface for creating ontologies and supports reasoners such as HermiT. HermiT checks whether the ontology is logically consistent and whether the defined relationships are valid.

In an academic organization, many entities are connected. Students enroll in courses, professors teach courses, departments offer courses, faculties contain departments, and students take examinations. This makes the academic domain suitable for ontology modeling.

3 Objectives

The main objectives of this project are:

- To design a comprehensive Academic Organization Ontology.

- To represent academic entities using OWL 2.0.
- To model students, professors, departments, faculties, and courses.
- To define relationships between academic individuals.
- To create object properties with appropriate domains and ranges.
- To create data properties for literal values such as names, IDs, dates, and codes.
- To add individuals representing real academic examples.
- To verify logical consistency using the HermiT Reasoner.
- To execute DL Queries for retrieving academic knowledge.
- To publish the complete ontology project on GitHub.

4 Tools and Technologies

Tool / Technology	Purpose
Protégé 5.x	Ontology development environment
OWL 2.0	Ontology representation language
HermiT Reasoner	Consistency checking and reasoning
DL Query	Knowledge retrieval from ontology
GitHub	Project repository and submission
LaTeX	Professional report writing

Table 1: Tools and technologies used in the project

5 Ontology Overview

The developed ontology is named **AcademicOrganizationOntology**. It represents a university-level academic structure and includes major academic and administrative entities.

The ontology contains:

- **56 Classes**
- **45 Object Properties**
- **41 Data Properties**
- **82 Individuals**

- **854 Axioms**

The ontology IRI is:

[http://www.semanticweb.org/mahamuda/ontologies/2026/
AcademicOrganizationOntology](http://www.semanticweb.org/mahamuda/ontologies/2026/AcademicOrganizationOntology)

This ontology provides a semantic model of an academic organization by describing how people, programs, departments, courses, and academic services are related.

6 Class Hierarchy

The class hierarchy is one of the most important parts of the ontology. Classes represent the main concepts of the domain. In this ontology, classes are organized according to the academic environment.

6.1 Main Classes

The main classes include:

- AcademicOrganization
- Faculty
- Department
- AcademicProgram
- Course
- Semester
- Classroom
- Laboratory
- Library
- ResearchCenter
- Examination
- Assignment
- Result
- Attendance

- Scholarship
- Certificate
- Event
- Hostel
- Transport
- Person

6.2 Person Class

The class **Person** represents human actors in the academic organization. It has several subclasses.

- Student
- UndergraduateStudent
- GraduateStudent
- PhDStudent
- Professor
- AssistantProfessor
- AssociateProfessor
- Lecturer
- TeachingAssistant
- Staff
- Administrator
- Dean
- Registrar
- ViceChancellor

6.3 Academic Classes

Academic classes describe learning and teaching components.

- Course represents a subject taught in the institution.
- AcademicProgram represents degree programs.
- Semester represents academic terms.
- Examination represents exams.
- Assignment represents coursework.
- Result represents academic performance.
- Attendance represents student participation.

7 Object Properties

Object properties define relationships between individuals. They connect one individual to another individual.

Object Property	Domain	Range
belongsToDepartment	Person	Department
belongsToFaculty	Department	Faculty
belongsToProgram	Student	AcademicProgram
enrolledIn	Student	Course
teaches	Professor	Course
takesCourse	Student	Course
offersCourse	Department	Course
conductsExam	Professor	Examination
takesExam	Student	Examination
assignedToClassroom	Course	Classroom
usesLibrary	Student	Library
usesLaboratory	Student	Laboratory
worksIn	Staff	Department
hasAdvisor	Student	Professor
supervisesStudent	Professor	Student
managesDepartment	Administrator	Department
headsDepartment	Professor	Department
belongsToUniversity	Department	AcademicOrganization

Object Property	Domain	Range
hasResult	Student	Result
hasAttendance	Student	Attendance
hasAssignment	Course	Assignment
submitsAssignment	Student	Assignment
evaluatedBy	Examination	Professor
gradedBy	Assignment	Professor
issuesCertificate	Registrar	Certificate
organizesEvent	Department	Event
participatesIn	Student	Event
providesScholarship	AcademicOrganization	Scholarship
locatedIn	Classroom	AcademicOrganization
reportsTo	Staff	Administrator
registeredIn	Student	AcademicProgram
hasSemester	AcademicProgram	Semester
hasProgram	Department	AcademicProgram
hasInstructor	Course	Professor
scheduledIn	Course	Timetable
memberOfResearchGroup	Professor	ResearchCenter
participatesInProject	Student	ResearchCenter
writesPublication	Professor	Publication
organizedBy	Event	Department
usesTransport	Student	Transport
livesInHostel	Student	Hostel
hasOffice	Professor	Office
hasTimetable	Course	Timetable
hasLibraryCard	Student	Library
receivesScholarship	Student	Scholarship

Table 2: Object properties with domain and range

These object properties provide the semantic structure of the ontology. For example, the property **enrolledIn** connects a student with a course, while **teaches** connects a professor with a course.

8 Data Properties

Data properties connect individuals with literal values such as strings, numbers, dates, and identifiers.

Data Property	Domain	Range
studentID	Student	xsd:string
teacherID	Professor	xsd:string
employeeID	Staff	xsd:string
fullName	Person	xsd:string
email	Person	xsd:string
phoneNumber	Person	xsd:string
gender	Person	xsd:string
age	Person	xsd:integer
cgpa	Student	xsd:decimal
credits	Course	xsd:integer
semesterNo	Semester	xsd:integer
roomNumber	Classroom	xsd:string
courseCode	Course	xsd:string
courseTitle	Course	xsd:string
departmentName	Department	xsd:string
facultyName	Faculty	xsd:string
programName	AcademicProgram	xsd:string
examDate	Examination	xsd:dateTime
examTime	Examination	xsd:string
assignmentMarks	Assignment	xsd:decimal
resultGrade	Result	xsd:string
attendancePercentage	Attendance	xsd:decimal
joiningDate	Staff	xsd:date
salary	Staff	xsd:decimal
libraryCardID	Library	xsd:string
eventDate	Event	xsd:date
certificateNo	Certificate	xsd:string
programDuration	AcademicProgram	xsd:string
hostelName	Hostel	xsd:string
transportRoute	Transport	xsd:string
officeNumber	Office	xsd:string
publicationTitle	Publication	xsd:string
researchTitle	ResearchCenter	xsd:string
scholarshipAmount	Scholarship	xsd:decimal
enrollmentDate	Enrollment	xsd:date
timetableCode	Timetable	xsd:string
labName	Laboratory	xsd:string

Data Property	Domain	Range
libraryName	Library	xsd:string
examRoom	Examination	xsd:string
certificateDate	Certificate	xsd:date
graduationDate	Graduation	xsd:date

Table 3: Data properties with domain and range

These data properties allow the ontology to store descriptive values. For example, a student can have a student ID, name, email address, CGPA, and phone number.

9 Individuals

Individuals are instances of classes. They represent concrete examples in the ontology.

Class	Sample Individuals
Student	Ali, Sara, Ahmed, Fatima, Usman, Ayesha
Professor	ProfAhmed, ProfFatima, ProfAli, ProfSara
Department	ComputerScienceDepartment, MathematicsDepartment
Faculty	EngineeringFaculty, ScienceFaculty
Course	KnowledgeRepresentation, ArtificialIntelligence, MachineLearning
AcademicProgram	BSCS, BSMath, MSAI, MSMathematics
Classroom	Room101, Room102, Room201
Library	CentralLibrary, DigitalLibrary
Laboratory	ResearchLab, ComputerLab
Examination	ExamAI, ExamMath, MidtermExam
Assignment	Assignment1, Assignment2
ResearchCenter	AIResearchCenter, DataScienceCenter
Scholarship	MeritScholarship, NeedBasedScholarship

Table 4: Sample individuals in the ontology

The ontology contains 82 individuals. These individuals help demonstrate how academic entities are connected through object and data property assertions.

10 Property Assertions

Property assertions are used to connect individuals with other individuals or literal values.

10.1 Example: Ali

- Ali is an instance of Student.
- Ali enrolledIn KnowledgeRepresentation.
- Ali enrolledIn ArtificialIntelligence.
- Ali belongsToDepartment ComputerScienceDepartment.
- Ali belongsToProgram MSAI.
- Ali hasAdvisor ProfAhmed.
- Ali usesLibrary CentralLibrary.
- Ali usesLaboratory ResearchLab.
- Ali takesExam ExamAI.

10.2 Example: Sara

- Sara is an instance of Student.
- Sara enrolledIn MachineLearning.
- Sara belongsToDepartment MathematicsDepartment.
- Sara belongsToProgram MSMathematics.
- Sara hasAdvisor ProfFatima.
- Sara usesLibrary CentralLibrary.
- Sara takesExam ExamMath.

10.3 Example: ProfAhmed

- ProfAhmed is an instance of Professor.
- ProfAhmed teaches KnowledgeRepresentation.
- ProfAhmed teaches ArtificialIntelligence.
- ProfAhmed supervisesStudent Ali.
- ProfAhmed worksIn ComputerScienceDepartment.
- ProfAhmed conductsExam ExamAI.

10.4 Data Property Examples

- Ali fullName “Ali”
- Ali studentID “ST001”
- Ali cgpa 3.75
- Sara fullName “Sara”
- Sara studentID “ST002”
- ProfAhmed fullName “Professor Ahmed”
- KnowledgeRepresentation courseCode “KR601”
- KnowledgeRepresentation credits 6

11 Reasoning Using HermiT

Reasoning is an important part of ontology development. The HermiT Reasoner was used to check the logical consistency of the ontology.

The reasoner verified:

- Class hierarchy
- Object property hierarchy
- Data property hierarchy
- Class assertions
- Object property assertions
- Same individuals

After running HermiT, no inconsistency was found. This confirms that the ontology is logically consistent. Reasoning also helps infer additional knowledge based on the defined domain and range restrictions.

12 DL Queries

DL Queries were used to retrieve information from the ontology.

12.1 Query 1: Students

Student

This query retrieves all individuals that belong to the class Student.

Expected result examples:

- Ali
- Sara
- Ahmed
- Fatima

12.2 Query 2: Professors

Professor

This query retrieves all professors in the ontology.

Expected result examples:

- ProfAhmed
- ProfFatima
- ProfAli

12.3 Query 3: Courses

Course

This query retrieves all course individuals.

Expected result examples:

- KnowledgeRepresentation
- ArtificialIntelligence
- MachineLearning
- DatabaseSystems

12.4 Query 4: Students Enrolled in Courses

Student and (enrolledIn some Course)

This query retrieves students who are enrolled in at least one course.

12.5 Query 5: Professors Teaching Courses

Professor and (teaches some Course)

This query retrieves professors who teach at least one course.

12.6 Query 6: Students Having Advisors

Student and (hasAdvisor some Professor)

This query retrieves students who have a professor assigned as advisor.

12.7 Query 7: Departments Offering Courses

Department and (offersCourse some Course)

This query retrieves departments that offer courses.

13 GitHub Repository

The complete project has been uploaded to GitHub. The repository includes the OWL ontology file, README documentation, and project report.

- AcademicOrganizationOntology.owl
- Project_Report.pdf
- README.md

GitHub Repository:

<https://github.com/mahamudaakter60/Academic-Organization-Ontology>

14 Results

The Academic Organization Ontology was successfully developed and tested. The ontology contains a large number of classes, object properties, data properties, and individuals. The use of HermiT Reasoner confirmed that the ontology is logically consistent.

The DL Queries successfully retrieve relevant academic information such as students, professors, courses, departments, and relationships between them. The ontology demonstrates how Semantic Web technologies can be used to model an academic organization.

15 Conclusion

This project successfully developed an Academic Organization Ontology using OWL 2.0 in Protégé. The ontology models the main structure of a university environment, including students, professors, departments, faculties, courses, classrooms, examinations, libraries, research centers, and administrative entities.

The ontology includes 56 classes, 45 object properties, 41 data properties, and 82 individuals. HermiT Reasoner was used to verify the logical consistency of the ontology. The DL Queries also demonstrate that the ontology can retrieve useful academic knowledge.

Overall, this project shows how ontology engineering can support semantic representation of academic institutions and improve structured knowledge management in educational environments.

16 Future Work

In the future, this ontology can be extended by adding more complex academic rules, such as prerequisite courses, grading policies, exam schedules, student progression, and research collaboration networks. It can also be connected with real university datasets to support automated academic information systems.

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